Impact of Consumers Traceability Awareness on Blockchain Adoption in Supply Chains

Xiaopo Zhuo, Yajie Sun, and Shaorui Zhou

***Abstract*—Blockchain technology (BCT) has emerged as a promising solution for ensuring supply chain traceability. However, not all consumers have a comprehensive understanding of the benefits associated with BCT-enabled traceability. In this article, we investigate the impacts of consumer awareness on the adoption of BCT within a supply chain comprising a manufacturer and a retailer. We develop two distinct scenarios: Scenario B, where the supply chain traceability is managed via traditional digital systems, and Scenario E, where the supply chain traceability is managed via BCT-enabled systems. We introduce the concept of consumer traceability awareness level, representing the proportion of the consumer population that is knowledgeable about the advantages of these traceability technologies. The findings reveal that the adoption of BCT enhances the overall performance of the supply chain and makes it more sensitive to the consumer traceability awareness level. Nonetheless, the manufacturer consistently experiences advantages from BCT adoption, whereas the retailer's situation may deteriorate. In both scenarios, a low traceability awareness level prompts the retailer to target all consumers, whereas a high-traceability awareness level shifts its focus solely to the knowledgeable consumers. Intriguingly, the adoption of BCT shifts the retailer's inclination toward targeting the knowledgeable consumers rather than all consumers.**

***Managerial Relevance Statement*—First, supply chain managers can use our stylized model to measure the impact of consumer traceability awareness on blockchain technology (BCT) adoption in supply chains. Second, the supply chain managers can use our designed cost-sharing contract for the upstream and downstream supply chain members to ensure a win–win outcome. Third, the supply chain members are suggested to promote the benefits of BCT to raise the consumer traceability awareness level and trust level. Finally, some specific operational strategies are proposed for supply chain members to adopt BCT considering consumer traceability awareness. These strategies allow them to develop more profitable target strategies according to the requirements of different consumer groups, and avoid experiencing a decline in profits due to blind pricing. For example, we suggest that the retailer targets all consumers when the awareness level is low and shifts its focus solely to the knowledgeable consumers when the awareness level is high. Surprisingly, the retailer considering adopting BCT for managing traceability is suggested to target knowledgeable consumers rather than all consumers.**

***Index Terms*—Blockchain technology (BCT), consumer awareness, supply chain management, traceability.**

NOMENCLATURE

Parameters:

Q	Demand quantity.
x_i	Consumers' trust level, $x_i \in [0,1]$ and $i \in \{H,L\}$
ϕ	Consumer awareness level.
α	Transaction cost coefficient.
δ	Sensitivity coefficient of consumer utility on BCT investment level.
γ	Cost-sharing ratio.
k_q	Cost coefficient of quality investment.
k_e	Cost coefficient of BCT investment.
π_m	Profit of the manufacturer.
π_r	Profit of the retailer.
CW	Consumer welfare.
SW	Social welfare.

Decision variables:

w Wholesale price.
q Product quality.
P Retail price.
e Level of investments in BCT.

I. INTRODUCTION

IN TODAY'S interconnected global supply chains, a finished product may involve multiple tiers of suppliers, thousands of globally sourced ingredients and raw materials, as well as numerous transitions between manufacturing and logistics' partners. In the absence of robust traceability, a quality issue at any stage of the production process can trigger a series of problems that can only be remedied at significant costs to brand owners, potentially leading to reputational damage. The impact will become more pronounced when consumers lose faith in the brand and stop buying its products because consumers nowadays increasingly view verifiable traceability evidence as a crucial standard for evaluating the quality and safety of a product [1]. Therefore, enterprises consider traceability a more prominent priority than before and traceability stands out as a critical performance metric for managing global supply chains.

Received 4 May 2024; revised 9 July 2024, 14 October 2024, 18 December 2024, and 15 February 2025; accepted 13 March 2025. Date of publication 18 March 2025; date of current version 4 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 72371252, Grant 72271251, and Grant 72171239, in part by the Basic and Applied Basic Research Project of Guangzhou under Grant SL2024A04J01795, and in part by the Natural Science Foundation of Guangdong Province, China, under Grant 2023A1515010683. Review of this article was arranged by Department Editor C. Amrit. *(Xiaopo Zhuo and Yajie Sun are co-first authors.) (Corresponding author: Shaorui Zhou.)*

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Digital Object Identifier 10.1109/TEM.2025.3552437

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Enterprises usually integrate traceability software into their enterprise resource planning (ERP) systems to enhance traceability. For instance, Software Advice showcases a comprehensive array of 74 traceability software solutions tailored for the food sector, including CSB-System, which encompasses a diverse range of functionalities, including process planning, computer-integrated manufacturing, automated data capture, and equipment control and maintenance [2]. A system like this can assist enterprises in capturing traceability data, such as the consumption of raw materials and semifinished products, by utilizing barcode or radio frequency identification (RFID) technology. It also enables customers to generate batch histories of the raw materials and tooling utilized for each output batch produced. Nevertheless, this architecture is commonly supervised by a centralized entity, a characteristic that exposes the system to vulnerabilities, such as data tampering, overwriting, and tag cloning [3]. In addition, there are several challenges associated with adopting an ERP-enabled traceability system. These include ensuring the reliability, security, and safety of data within the system, addressing the complexity and expense of reliable transactions, and ensuring compatibility between the system and business processes [4].

Recently, blockchain technology (BCT) has been increasingly recognized as a promising solution for ensuring traceability in supply chain management [5]. In contrast to ERP systems that depend on a centralized database, BCT operates on a distributed ledger architecture. This implies that each transaction executed through BCT is fortified by cryptography, ensuring verification, immutability, and resistance to tampering. Furthermore, BCT facilitates the establishment of digitized smart contracts that streamline processes, such as payments, effectively obviating the necessity for the third-party verification of compliance and diminishing transaction costs within supply chains. Therefore, BCT can effectively solve the complex business challenges encountered by ERP-enabled systems and, consequently, has become an increasingly popular technology adopted across various sectors to ensure traceability. As an example, the Walmart China Blockchain Traceability Platform was introduced in 2019, and subsequently, 23 distinct product lines were subjected to testing and launched on the platform [4]. To establish trustworthy and transparent consumer brands, MTR Foods Limited utilizes TraceX's Blockchain traceability solutions to trace the source of the Byadgi chili used in their products [6].[1]

To effectively adopt a BCT-enabled traceability system, one of the key challenges is to ensure that consumers are knowledgeable about the superiorities of BCT in enhancing traceability [4]. A high level of consumer awareness signifies that a substantial portion of the consumer population possesses knowledge about the advantages of employing BCT for traceability purposes. In general, consumers tend to prioritize price over other factors, particularly in developing regions where incomes are still low. This makes products with traceable origins less competitive due to the additional costs associated with adopting a BCT-enabled system. Differently, consumers who are knowledgeable about the benefits of BCT in traceability have a high evaluation of product quality and become less price sensitive than those who are unknowledgeable. As a result, how many consumers are knowledgeable about the benefits of BCT in traceability has a significant impact on both demand and the adoption of BCT-enabled systems in supply chains [7]. The aforementioned practices of BCT adoption in supply chains motivate us to pose the subsequent research questions.

1) Should a supply chain undertake investment in BCT to manage traceability?
2) What are the impacts of consumer traceability awareness on the adoption of BCT in supply chains?
3) What are the impacts of BCT adoption on supply chain members' profits and consumer welfare?

To address these questions, we consider a supply chain composed of an upstream manufacturer and a downstream retailer. We develop two distinct scenarios: scenario B, where the supply chain members deploy traditional ERP-enabled systems for traceability, and scenario E, where the supply chain members deploy BCT-enabled systems for traceability. The cost of adopting BCT is shared between the manufacturer and the retailer via a cost-sharing contract. The level of consumer traceability awareness, characterized as the proportion of the consumer population possessing knowledge about the advantages associated with these traceability technologies, constitutes a pivotal variable within the model. Consumers who are aware of the advantages of these traceability technologies (hereafter referred to as "knowledgeable consumers") tend to trust the quality of products more and are more likely to make purchases compared with those who lack this knowledge. Hence, the retailer can target solely the knowledgeable consumers or all consumers. Our focus lies in examining how the level of consumer awareness influences the supply chain members' decisions regarding BCT adoption for supply chain traceability.

We present a summary of our key findings in response to the research questions. First, we examine the impact of BCT adoption and find that it enhances the overall performance of the supply chain. However, while the manufacturer consistently benefits from BCT adoption, the retailer may face challenges, especially when the shared BCT investment cost increases. Therefore, a feasible cost-sharing contract could be established to enhance collaboration between the manufacturer and retailer in terms of BCT investment. Moreover, adopting BCT tends to shift the retailer's focus toward knowledgeable consumers, rather than the entire consumer base. Second, we conduct an analysis of the influence exerted by consumer awareness. Our analysis reveals that the retailer targets knowledgeable consumers when the awareness level is high. On the contrary, when the awareness level decreases, the retailer expands its focus to encompass all consumers. A high awareness level naturally corresponds to increased BCT investment, greater profit for the manufacturer and retailer, and improved supply chain performance. Third, we investigate the impact of BCT adoption on consumer welfare and demonstrate its positive influence when the targeted consumer group is identical. Interestingly, we find that BCT adoption could negatively impact consumer welfare primarily because it pushes up the retail price and reduces demand.

[1]For additional examples, visit TraceX Technologies' official website at https://tracextech.com/

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The contributions of this article are threefold. First, we augment the existing research by analytically exploring the impact of consumer traceability awareness on BCT adoption in supply chains. Consumer awareness, especially in the context of emerging technologies, such as BCT, has been notably underexplored in prior studies. Second, our model encapsulates specific aspects of BCT adoption, including the enhancement of trust levels and the elimination of transaction costs. This allows our model to more precisely mirror the business process of adopting BCT in supply chains. Third, we develop a stylize model for the supply chain managers to measure the impact of consumer traceability awareness on BCT adoption in supply chains and the analytical results provide several operational strategies. For example, a feasible cost-sharing contact is designed to ensure all supply chain members benefit from BCT adoption. In addition, we suggest the supply chain members to promote the benefits of BCT to raise the consumer traceability awareness level and trust level.

The rest of this article is organized as follows. Section II offers an overview of related studies. Section III delineates the fundamental concepts of the model. Section IV analyzes the impact of BCT adoption on the decisions and performance of supply chain members. Section V extends the core models, as well as discusses consumer welfare and social welfare. Finally, Section VI concludes this article.

II. Literature Review

Our work is related to three fundamental research streams:

1) the application of BCT in supply chains;
2) supply chain traceability management;
3) consumer awareness of emerging technology.

Next, we briefly review the relevant studies within these domains.

A. Application of BCT in Supply Chains

The first related stream is research on BCT application in supply chains, which has seen significant growth and offers promising opportunities. Babich and Hilary [8] identify key strengths and corresponding limitations in the application of BCT within operations management. Wong et al. [9] identify the psychological and systems-related barriers to adopting BCT and find nonlinear relationships of the barriers toward BCT adoption in operations management. Supply chain finance (SCF) has emerged as a vibrant and expansive application area of BCT in recent years. Du et al. [10] contribute to this domain by introducing a novel SCF platform that utilizes BCT to oversee the entire process, addressing issues of mistrust, enhancing capital flow efficiency, and delivering improved financial services. Dong et al. [11] undertake a comparison between a conventional system marked by restricted visibility and a blockchain-enabled system characterized by comprehensive transparency. Furthermore, the application of BCT has garnered increasing attention in other fields, including healthcare [12], agri-food [13], E-commerce [14], patents [15], green products [16], and sustainable supply chains [17].

Recently, applying BCT to manage supply chain traceability has received increasing attention. Azzi et al. [18] identify the benefits and challenges of applying BCT in supply chains to create a reliable, transparent, authentic, and secure system. Zhu and Kouhizadeh [19] investigate the contribution of BCT in addressing information challenges in product management by applying BCT to help product deletion and rationalization decisions. Hastig and Sodhi [20] examine the cobalt mining and pharmaceutical sectors, delving into these industries to uncover the requisites and pivotal elements that pave the way for the successful adoption of BCT in supply chain traceability. Guo et al. [21] explore the impacts of BCT adoption on the information disclosure games over the environmental efforts in the fashion industry, by considering two different information disclosure strategies for the retailer to declare the environmental quality to consumers, namely, ecolabeling or BCT adoption. Sunny et al. [22] offer a comprehensive survey of diverse blockchain-based traceability solutions discussed in the existing studies. Their work includes a proof of concept in the context of a cold chain, illustrating how a blockchain-based traceability solution can effectively augment supply chain traceability. Biswas et al. [3] develop a game-theoretic model to explore the tradeoffs between traceability and sustainability in BCT implications within a supply chain consisting of a manufacturer and a retailer. They reveal that a high level of distrust may deter firms from adopting BCT.

B. Supply Chain Traceability Management

Several studies emphasize the management of supply chain traceability, with a specific focus on safety-critical sectors, such as agri-food and pharmaceuticals. For example, Dabbene et al. [23] conduct a comprehensive review of food supply chain management and explore the impact of traceability and discuss various criteria for evaluating the performance of traceability systems. Liu et al. [24] propose a smart tracking and tracing platform based on BCT and Internet of Things and offer a decentralized traceability solution in the pharmaceutical supply chain. Typically, enterprises integrate traceability software into their ERP systems to enhance traceability. For example, Savino et al. [25] propose a framework through a case study; they integrate the supply chain operations reference model in the ERP system for an RFID-enabled supply chain to achieve greater product traceability. Moreover, ERP systems have evolved via the integration of new modules with enhanced capabilities. Such capabilities include new technological advancements, such as blockchain-based ERP systems. For instance, Perera et al. [26] examine the application of BCT-driven ERP systems and investigate the benefits and challenges for the supply chain in achieving transparency and authenticity.

Furthermore, the interaction between traceability and other supply chain characteristics, such as coordination, transparency, and resilience, is discussed in the existing studies. For instance, Shou et al. [27] explore the realm of operational innovation within supply chains through the creation of a system that harmonizes traceability and supply chain coordination. Their analysis

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highlights the potential for companies to attain operational innovation by effectively aligning the concepts of traceability and supply chain coordination. Inman and Green [28] examine the correlations among complexity, traceability, and transparency of adverse events, using covariance-based structural equation models. Their results indicate contradictive findings that there might be positive impacts of supply chain complexity on traceability and transparency. Manzoor et al. [29] conduct a systematic review and identify several enablers and barriers that facilitate or hinder the adoption of BCT in enhancing supply chain resilience. They deliver a critical insight indicating that promoting BCT adoption inherently benefits traceability and, hence, enhances resilience.

C. Consumer Awareness of Emerging Technologies

A limited number of studies consider the element of consumer awareness within the context of adopting emerging technologies. Claudy et al. [30] conduct an empirical study within the Republic of Ireland, aiming to assess consumer awareness concerning the adoption of microgeneration technologies. Their findings indicate substantial variations in awareness levels across diverse technologies and customer segments. Cottrill and Thakuriah [31] examine consumer privacy awareness and concerns related to location-sensing technologies. Through a survey instrument, they assess factors, such as knowledge, risk perception, willingness to trade privacy for benefits, and overall awareness. Wang et al. [32] investigate the correlation between consumer awareness regarding the advantages of recently developed vaccines and the influence it exerts on government subsidy programs. They underscore the significance of governmental efforts to disseminate information about vaccine benefits, as it effectively raises consumer awareness.

In recent, a rising topic concerning consumer awareness of traceability has gained limited attention within the context of adopting BCT in managing supply chain traceability. By considering the average value of the traceability awareness of consumers, Fan et al. [7] study the optimal pricing strategies within a supply chain consisting of a supplier, a manufacturer, and a retailer. They emphasize the notable impact of consumer traceability awareness on supply chain decisions regarding the adoption of BCT. Employing a similar approach, Tan et al. [33] investigate the impact of consumer awareness of traceability on the implementation of BCT in a fresh product supply chain, which consists of an e-commerce platform and a retailer. Their results reveal that BCT is implemented only when consumer traceability awareness is high. Distinctively, Sun et al. [34] construct a game-theoretic model that captures the specific characteristics of consumer awareness toward drug traceability in the anticounterfeiting context. They show the significant impact of consumer traceability awareness on BCT adoption of brand and generic pharmaceutical firms in a specific competitive environment.

D. Summary of the Literature

This study differs from the existing literature in several key aspects. First, the literature in the stream of the application of BCT in supply chains mainly focuses on identifying the benefits, challenges, and research opportunities [8], [18], [35] and investigating the impacts of applying BCT to manage supply chain traceability [3], [21]. We extend the existing research and explore the impacts of BCT adoption by taking into account critical factors, such as consumers' traceability awareness. Our results augment the existing research and demonstrate that BCT adoption enhances the overall performance of the supply chain and makes it more sensitive to the consumer awareness level.

Second, there are several studies that emphasize the management of supply chain traceability with a focus on safety-critical sectors [23], [24] and through ERP systems [25], [26]. We conceptualize two typical traceability management systems: ERP-enabled and BCT-enabled systems. Notably, we incorporate the unique attributes of these systems into our model, including factors, such as consumer trust levels and transaction costs. Our results reveal that the manufacturer consistently experiences advantages from adopting BCT-enabled systems, whereas the retailer's situation may deteriorate. We show that a feasible cost-sharing contract could be designed to enhance collaboration within supply chains in terms of BCT investment.

Third, compared with the previous studies that consider the average value of the traceability awareness of consumers [7], [33], we take a distinctive approach by focusing on two distinct consumer groups characterized by varying degrees of awareness regarding the benefits associated with BCT-enabled traceability. This divergence in awareness levels subsequently results in differing levels of trust in the quality of products within supply chains. We extend the literature and present intriguing findings that consumer traceability awareness significantly shapes BCT adoption and performance within supply chains. For instance, adopting BCT shifts the retailer's inclination toward targeting the knowledgeable consumer group rather than the entire consumer base.

By concentrating on these novel aspects, we offer valuable insights into the crucial role of consumer traceability awareness and the wider implications of incorporating BCT into supply chain operations. The comparison between this study and the literature is summarized in Table I.

III. Model

We consider a supply chain system comprising a manufacturer and a retailer, in which the retailer procures products from the manufacturer at a wholesale price denoted as w and subsequently vends them to consumers at a retail price represented as P. The product quality made by the manufacturer is denoted as q, incurring an investment cost of $\frac{k_q q^2}{2}$, where k_q represents the cost coefficient. Such a classical quadratic convex form is widely used in the literature [3], [5], which means that, for a product of higher quality, each unit will cost the manufacturer more in terms of production.

To achieve traceability in the supply chain, a traditional ERP-enabled system or a BCT-enabled system is generally adopted. However, it is crucial to acknowledge that not all consumers are well versed in these traceability technologies. We define the consumer traceability awareness level as the proportion of the

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TABLE I
COMPARISON BETWEEN THIS STUDY AND THE LITERATURE

Article	Blockchain investor	How is BCT adoption considered in supply chains	Consumer trust	Consumer traceability awareness
[11]	Supply chain	Improve supply chain visibility		
[36]	Supply chain	Enlarge market demand; Improve credit value		
[21]	Retailer	Disclose environmental quality; Eliminate inspections; Devote advertisement costs		
[3]	Downstream firm	Improve product traceability; Eliminate transaction costs; Generate energy consumption	√	
[7]	Supply chain	Satisfy the traceability requirement of consumers		√
[33]	Platform	Disclose actual quality level; Satisfy the traceability requirement of consumers		√
[34]	Manufacturer	Anti-counterfeiting; Eliminate consumers' detection costs		√
This study	Supply chain	Eliminate transaction costs; Improve consumers' willingness to pay and trust level	√	√

consumer population that possesses knowledge about the advantages associated with these traceability technologies. Hence, we can distinguish between two types of consumers, denoted by $i \in \{H, L\}$, and type-H consumers are knowledgeable about the advantages of these traceability technologies, whereas type-L consumers are not. Let $\phi \in [0, 1]$ represents the proportion of the population that consists of type-H consumers, and x_H denotes the trust level in product quality for these consumers. As a result, the proportion of the population that consists of type-L consumers is $1 - \phi$, and they have a trust level in product quality denoted by x_L. A high-traceability awareness level indicates that the proportion of the population that consists of type-H consumers is large, namely, a high value of ϕ. In addition, a high value of $x_i \in (0, 1]$ indicates that consumers have a great degree of trust in product quality. Intuitively, type-H consumers exhibit a greater degree of trust in the quality of products and are inclined to make purchases in comparison with type-L consumers, which implies that $x_H > x_L$. The net utility of a consumer for buying a product is defined as $\theta q x_i - P$, where θ represents the marginal valuation for the product and is uniformly distributed on the interval $[0, 1]$ for both types of consumers. Intuitively, a consumer will opt not to purchase the product if the net utility turns out to be negative, meaning $\theta < \frac{P}{qx_i}$. Hence, the proportion of type-i consumers who purchase products is $(1 - \frac{P}{qx_i})^+$. Recall that the proportion of the type-H and type-L consumers is ϕ and $1 - \phi$, respectively. Hence, the total demand quantity, which includes both type-H and type-L consumers, is specified as $Q(P) = \phi(1 - \frac{P}{qx_H})^+ + (1 - \phi)(1 - \frac{P}{qx_L})^+$.

To examine supply chain members' incentives for achieving supply chain traceability via BCT, we develop two distinct scenarios: scenario B, where supply chain members manage traceability via traditional ERP-enabled systems, and scenario E, where the supply chain members manage traceability via BCT-enabled systems. Adopting BCT in the supply chain can lead to two primary advantages. First, it ensures a high level of product traceability, enhancing consumers' willingness to pay. Second, it eliminates the transaction costs associated with intermediaries, supply chain coordination, and reliable transactions. Through the adoption of BCT, all information recorded by supply chain members becomes traceable, and payments can occur without requiring intermediaries for compliance verification. However, adopting BCT also introduces certain business drawbacks. One notable example is the substantial investment cost of BCT, which is associated with achieving supply chain traceability and can potentially influence consumers' willingness to pay. Next, we elucidate the profit function pertaining to the supply chain members and outline the sequence of events within the two scenarios. To simplify cross referencing, we provide a summary of the notations and definitions in Nomenclature. In both scenarios, the parameters are annotated with superscripts H and A to denote that the retailer targets the knowledgeable consumers or all consumers, respectively.

A. Scenario B

In scenario B, supply chain members manage traceability through traditional ERP-enabled systems, which are considered less effective than BCT-enabled systems [20]. Consequently, a consumer's net utility is influenced by both product quality and retail price, with the demand quantity defined as follows:

$$Q(P) = \phi\left(1 - \frac{P}{qx_H}\right)^+ + (1 - \phi)\left(1 - \frac{P}{qx_L}\right)^+. \quad (1)$$

The manufacturer sets the wholesale price and product quality, while the retailer determines the retail price. As a result, we can delineate the specific problems faced by both the manufacturer and the retailer as follows:

$$\max_{w,q} \pi_m = Qw - \frac{k_q q^2}{2} \quad (2)$$

$$\max_{P} \left\{ \sup_{P > qx_L} \pi_r(P), \sup_{P \le qx_L} \pi_r(P) \right\}. \quad (3)$$

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In (3), $\pi_r(P)$ represents the retailer's profit, defined as $\pi_r(P) = Q(P - w(1+\alpha))$. Here, αw represents the marginal transaction cost incurred by the retailer for using ERP-enabled systems to finalize the payments and ensure reliable transactions due to the lack of transparency regarding contract compliance [3], [37], with α being the transaction cost coefficient. In scenario E, the adoption of BCT eliminates this transaction cost. Note that $x_H > x_L$, and a consumer will only purchase the products if the net utility is positive. When $P > qx_L$, only type-H consumers (i.e., knowledgeable consumers) have a positive utility. Therefore, the retailer targets knowledgeable consumers, with the demand function defined as $Q(P) = \phi(1 - \frac{P}{qx_H})$. However, when $P \le qx_L$, both type-*H* and type-*L* consumers (i.e., all consumers) have a positive utility. In this case, the retailer targets all consumers, with the demand function defined as $Q(P) = \phi(1 - \frac{P}{qx_H})^+ + (1-\phi)(1 - \frac{P}{qx_L})^+$.

The sequence of events is straightforward. The sequence of events unfolds as follows. Initially, the manufacturer simultaneously sets the wholesale price and product quality level to maximize its gross profit. Subsequently, the retailer sets the retail price, targeting either all consumers or solely knowledgeable consumers, in an effort to maximize its gross profit. Using a backward induction approach, the equilibrium is inferred and detailed in Table A-I (see Appendix A for a detailed derivation).

B. Scenario E

In scenario E, the supply chain members invest in BCT to manage supply chain traceability. Traceability is inherently valuable for consumers, and the improved supply chain traceability can positively affect consumers' willingness to pay and, consequently, the consumer utility. Intuitively, the type-H consumers benefit more from BCT adoption than the type-L consumers due to their higher level of traceability awareness. To capture the positive effect of BCT adoption, we rewrite a consumer's net utility as $\theta(q+\delta e)\bar{x}_i - P$, in which e represents the investment level of BCT and δ is the sensitivity coefficient. In practice, BCT adoption necessitates robust Internet infrastructure (e.g., platforms, cryptocurrency development, and smart contract engineering) and a fundamental transformation of enterprise operations (e.g., finalizing transactions, exchanging money, and writing contracts). Hence, a higher investment level of BCT results in better supply chain traceability from product origin to distribution and, consequently, higher willingness to pay and utility of consumers. In addition, the trust level of type-H consumers, who are knowledgeable about the benefits of blockchain on traceability, will improve, as indicated by $\bar{x}_H > x_H$. However, the trust level of type-L consumers, who are unknowledgeable about the advantages of these traceability technologies, remains unchanged, as indicated by $\bar{x}_L = x_L$. Hence, considering BCT adoption in the supply chain, the demand quantity is specified as follows:

$$Q(P,e) = \phi\left(1 - \frac{P}{(q+\delta e)\,\bar{x}_H}\right)^+ + (1-\phi)\left(1 - \frac{P}{(q+\delta e)\,x_L}\right)^+. \quad (4)$$

From the above demand function, we observe that a higher investment level can increase demand volume, encompassing both type-H and type-L consumers. Accordingly, we can formulate the respective problems of the manufacturer and retailer as follows:

$$\max_{w,q,e} \pi_m(q,e) = Qw - \frac{k_q q^2}{2} - \gamma I(e) \quad (5)$$

$$\max_{P,e} \left\{ \sup_{P>(q+\delta e)x_L} \pi_r(P,e), \sup_{P\le(q+\delta e)x_L} \pi_r(P,e) \right\}. \quad (6)$$

The adoption of BCT-enabled systems can improve the business efficiency of the supply chain, thereby eliminating the retailer's transaction costs. In (6), $\pi_r(P,e)$ is the profit function of the retailer, which is specified as $\pi_r(P,e) = Q(P-w) - (1-\gamma)I(e)$. The investment cost of adopting the BCT-enabled systems is given as $I(e) = \frac{k_e e^2}{2}$ in which k_e is the cost coefficient. The BCT adoption can lead to better supply chain traceability from product origin to distribution. Hence, the BCT investment cost is shared by the manufacturer and the retailer [36], denoted by the cost-sharing ratio γ. Hence, when $P > (q+\delta e)x_L$, the retailer targets knowledgeable consumers, with the demand function defined as $Q(P,e) = \phi(1 - \frac{P}{(q+\delta e)\bar{x}_H})$. However, when $P \le (q+\delta e)x_L$, the retailer targets all consumers, with the demand function defined as $Q(P,e) = \phi(1 - \frac{P}{(q+\delta e)\bar{x}_H}) + (1-\phi)(1 - \frac{P}{(q+\delta e)x_L})$.

In sum, BCT adoption brings the following differences in the model:

1) improve supply chain traceability and, consequently, increase the consumer utility to $\theta(q+\delta e)\bar{x}_i - P$, in which the type-$H$ consumers benefit more from BCT adoption than the type-L consumers, namely, $\bar{x}_H > x_H$ and $\bar{x}_L = x_L$;
2) eliminate the retailer's transaction cost, namely, $\alpha = 0$;
3) induce an investment cost $I(e) = \frac{k_e e^2}{2}$, which is shared by the supply chain members with a cost-sharing ratio γ.

The sequence of events unfolds as follows. Initially, the manufacturer and retailer collaborate to set a BCT investment level, with the aim of maximizing the overall supply chain profit. Subsequently, the manufacturer simultaneously sets a wholesale price and a quality level to maximize its gross profit. Finally, the retailer sets the retail price, targeting either all consumers or solely knowledgeable consumers, in an effort to maximize its gross profit. The equilibrium is determined using backward induction and is presented in Table A-I (see Appendix A for a detailed derivation). Note that a bar is appended to the parameters in scenario E to distinguish them from those in scenario B.

IV. Analysis

In this section, we initially examine how market factors, such as awareness level, influence the equilibrium in both scenarios. Subsequently, we address whether supply chain members should invest in BCT-enabled systems for traceability management. Finally, we investigate the impact of BCT on the performance of supply chain members, as well as consumer welfare.

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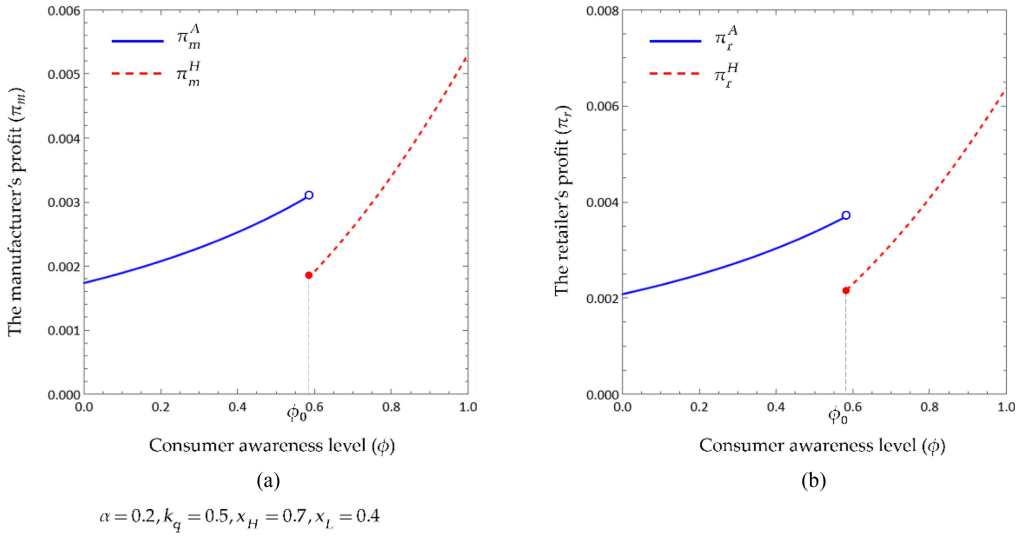


Fig. 1. Equilibrium profit of supply chain members in scenario B. (a) Impact of consumer awareness level on the manufacturer's profit. (b) Impact of consumer awareness level on the retailer's profit.

A. *Managing Traceability Through Traditional ERP-Enabled Systems*

When supply chain members manage traceability through traditional ERP-enabled systems, the impacts of key market factors, such as awareness level and consumer trust level, are summarized in the following proposition.

Proposition 1: In scenario B, the equilibriums P,w, q, π_m, and π_r increase in ϕ and x_i, but decrease in α and k_q regardless of the targeted consumer group.

According to Proposition 1, the equilibrium wholesale price, retail price, and product quality level increase with a rise in the awareness level. In other words, the equilibrium wholesale price, retail price, and product quality level increase when type-H consumers constitute a large proportion. This phenomenon can be attributed to type-H consumers' greater trust in product quality and lower price sensitivity, which makes improving product quality more advantageous [34]. As product quality improvement incurs additional production costs, it subsequently drives up the wholesale price and retail price. However, the equilibrium wholesale price, retail price, and product quality level decrease with the transaction cost coefficient (i.e., α) and cost coefficient of quality investment (i.e., k_q). A high value of α amplifies transaction cost, intensifying the double marginalization effect within the supply chain, leading to poorer performance of the manufacturer and the retailer. Differently, a high value of k_q drives the manufacturer to set a low level of product quality to reduce the production cost. Consequently, this leads to a low wholesale price and retail price, making the manufacturer and the retailer worse off.

Consumer trust in product quality positively affects the equilibrium wholesale price, retail price, and product quality [3], [38]. Intuitively, high consumer trust in product quality amplifies the positive impact of improved product quality on the manufacturer's utility. Hence, the manufacturer is motivated to establish a higher product quality level, despite the resultant increase in production costs and, subsequently, the wholesale and retail prices. Consequently, both the manufacturer and the retailer benefit from the improvement in consumer trust in product quality. This result suggests that supply chain members should focus on improving consumer trust in product quality. One possible approach is to adopt BCT-enabled systems to enhance the trust level of type-H consumers in product quality.

Next, we explore the equilibrium profits of both the manufacturer and the retailer, with the outcomes summarized in the following proposition.

Proposition 2: In scenario B, the retailer targets all consumers when $\phi \in [0, \phi_0)$, and then shifts its focus to the knowledgeable consumers when $\phi \in [\phi_0, 1]$, in which $\phi_0 = \min\{\frac{x_H}{4x_H - 4x_L}, 1\}$.

As stated in Proposition 2, the retailer may target either all consumers or the knowledgeable consumers, depending on the awareness level denoted as ϕ_0 [32]. To be specific, the retailer targets all consumers when the awareness level is low and then shifts its focus to the knowledgeable consumers when the awareness level is high (see Fig. 1 for a graphical illustration). When the awareness level is low, it implies that only a small proportion of the consumer population is aware of the advantages of ERP-enabled traceability. In this situation, providing a high-quality product at an elevated retail price would substantially impede the overall profitability of the supply chain members. Hence, the retailer opts to target all consumers to raise the demand volume and enhance its profitability. As the awareness level improves, namely, as the number of type-H consumers increases, the supply chain members will strive to offer a high-quality product along with an elevated retail price, aiming to increase their profit margins. However, the elevated retail price results in negative utility for type-L consumers, causing them to refrain from purchasing the products. Consequently, the retailer chooses to set a higher retail price and focus on the knowledgeable consumers.

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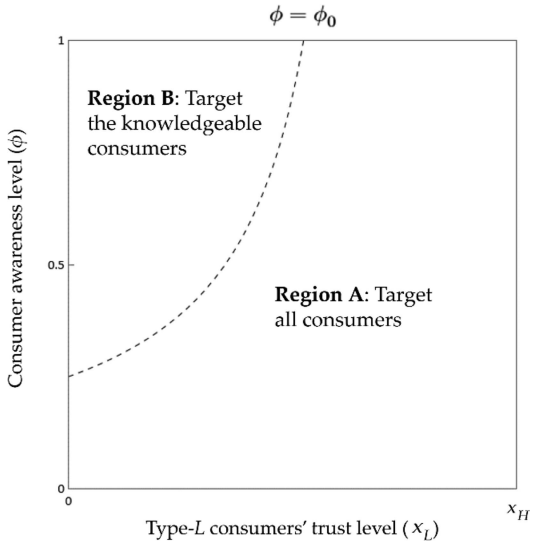


Fig. 2. Targeted consumer group in scenario B.

Fig. 1 illustrates that the equilibrium profit of both the manufacturer and the retailer increases with the awareness level, regardless of the targeted consumer group. This finding aligns with the results presented in Proposition 1. Nonetheless, it is crucial to note that increasing the awareness level does not always guarantee improved performance for the supply chain members. As shown in Fig. 1, targeting knowledgeable consumers may result in lower profits for the supply chain members compared with targeting all consumers. This outcome can be attributed to the following driving forces. Supply chain members gain a demand volume advantage when the retailer targets all consumers, but a profit margin advantage when the retailer targets knowledgeable consumers. As the awareness level increases, the retailer targets knowledgeable consumers in equilibrium due to the negative utility of type-L consumers. In this situation, the strengthening of the price margin advantage resulting from an increased retail price cannot compensate for the weakening of the demand volume advantage. The advantage of achieving a substantial price margin benefit by targeting knowledgeable consumers only becomes significant when the awareness level is substantially high. In such cases, choosing to target knowledgeable consumers can lead to improved performance for both the manufacturer and the retailer, surpassing the strategy of targeting all consumers.

Furthermore, our analysis extends to investigating the influence of consumers' trust levels in product quality on the equilibrium choices made by the supply chain members. As shown in Fig. 2, we observe that the retailer primarily targets all consumers in most cases. It is clear that the retailer will target all consumers when the awareness level is low, as the advantage of demand volume outweighs the price margin advantage in this situation. However, when the awareness level is high, the retailer targets knowledgeable consumers when the trust level of type-L consumers on product quality is low and targets all consumers when the trust level of type-L consumers on product quality is high. The driving forces behind this are as follows. When the awareness level is high, supply chain members are motivated to offer a high-quality product at a high retail price. In this situation, if the trust level of type-L consumers in product quality is high and their utility is positive, the retailer targets all consumers. Otherwise, the retailer targets knowledgeable consumers to leverage the price margin advantage.

B. Managing Traceability Through BCT-Enabled Systems

In scenario E, the manufacturer and retailer collaborate to adopt BCT-enabled systems for managing supply chain traceability. Adopting BCT eliminates the retailer's transaction costs and improves consumers' willingness to pay. In particular, it improves the trust level of type-H consumers in product quality, namely, $\bar{x}_H > x_H$, because they are knowledgeable about the benefits of BCT-enabled traceability. However, it incurs an investment cost that is shared between the manufacturer and retailer. We encapsulate these characteristics in scenario E and investigate the impact of BCT-enabled traceability systems. The results are summarized in the following proposition.

Proposition 3: In scenario E, regardless of the targeted consumer group, we have (a) P,w, q, e, π_m, and π_r increase in ϕ and x_i. (b) P,w, q, π_m, and π_r decrease in k_q; in addition, P,w, e, and π_m decrease in k_e, whereas $\frac{\partial \pi_r}{\partial k_e} < (>)0$ if $\gamma > (<)\frac{1}{3}$. (c) The manufacturer's profit always increases in δ, whereas the retailer's profit may increase or decrease in δ, namely, $\frac{\partial \pi_r}{\partial \delta} > (<)0$ if $\gamma > (<)\frac{1}{3}$.

Proposition 3 characterizes the impacts of key market factors on the equilibrium when adopting BCT-enabled systems. We demonstrate that supply chain members benefit from the increase in awareness level and consumer trust level [3], [7] [see Proposition 3(a)], which is identical to what is observed when adopting ERP-enabled systems. The BCT investment level also increases with the awareness level and consumer trust level. In this situation, investing in BCT can significantly improve consumer utility due to the increase in awareness level and consumer trust level. Consequently, supply chain members are incentivized to make substantial investments in BCT. This, in turn, leads to higher product quality, wholesale price, and retail price. Proposition 3(b) shows the impact of cost coefficients of quality investment (i.e., k_q) and BCT investment (i.e., k_e). A high value of k_q leads to low product quality, making the manufacturer and the retailer worse off. Interestingly, a high value of k_e always leads to poor performance of the manufacturer, whereas the retailer's performance can be positively or negatively affected. To be specific, the retailer's profit increases in k_e if $\gamma < \frac{1}{3}$. In this situation, the retailer bears most of the BCT investment cost (i.e., $\frac{(1-\gamma)k_e e^2}{2}$), which decreases significantly due to the reduction of BCT investment level e. Hence, a huge drop in BCT investment cost consequently benefits the retailer.

Proposition 3(c) shows the impact of sensitivity coefficient δ on the profit of supply chain members. To be specific, a high δ always benefits the manufacturer because BCT-enabled traceability is inherently valuable to consumers, and supply chain members are incentivized to make substantial investments in BCT to attract more demand when δ is high. The cost of BCT investment is shared by supply chain members, while the manufacturer can increase the wholesale price or improve

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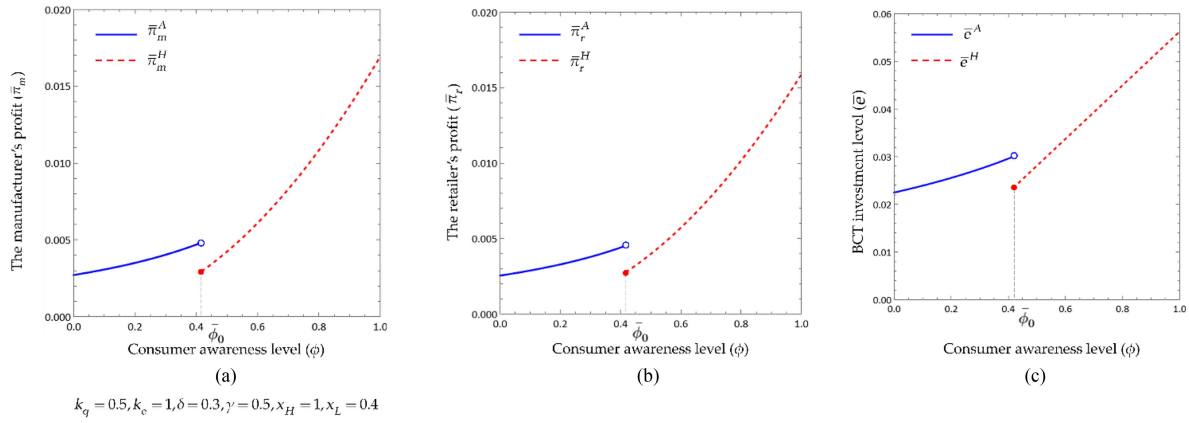


Fig. 3. Equilibrium profits and BCT investment level in scenario E. (a) Impact of consumer awareness level on the manufacturer's profit. (b) Impact of consumer awareness level on the retailer's profit. (c) Impact of consumer awareness level on BCT investment level.

product quality to cover its BCT investment cost. Interestingly, a high δ will reduce the retailer's profit when the cost-sharing coefficient is small (i.e., $\gamma < \frac{1}{3}$). In this situation, the retailer bears most of the BCT investment cost. The BCT investment cost is large because a high δ drives supply chain members to make substantial investments in BCT. The retailer cannot set a significantly high retail price because a high retail price will offset the positive effect of BCT on attracting demand. Consequently, the retailer's profit decreases with δ when it bears most of the BCT investment cost.

Next, we explore the equilibrium profit of both the manufacturer and the retailer in the context of managing traceability through BCT-enabled systems. The results are summarized in the following proposition.

Proposition 4: In scenario E, the retailer targets all consumers when $\phi \in [0, \bar{\phi}_0)$, and then shifts its focus to the knowledgeable consumers when $\phi \in [\bar{\phi}_0, 1]$ in which $\bar{\phi}_0 = \min\{\frac{\bar{x}_H}{4\bar{x}_H - 4x_L}, 1\}$.

Proposition 4 shows that, when managing traceability through BCT-enabled systems, a low awareness level prompts the retailer to target all consumers, while a high awareness level shifts its focus to knowledgeable consumers. This result is graphically presented in Fig. 3 , which is similar to the situation of managing traceability through traditional ERP-enabled systems. The fundamental driving forces are identical. A low awareness level prevents the retailer from setting a high retail price, leading it to target all consumers. Conversely, a high awareness level allows the retailer to set a high retail price to improve its profit margin, leading it to target knowledgeable consumers.

By comparing Propositions 2 and 3, we can easily see that the thresholds for the targeted consumer group, namely, ϕ_0 and $\bar{\phi}_0$, are different. By comparing the values of ϕ_0 and $\bar{\phi}_0$, we obtain the following proposition.

Proposition 5: The adoption of BCT makes the retailer more inclined to target knowledgeable consumers, namely, $\bar{\phi}_0 < \phi_0$.

Proposition 5 presents an interesting finding that the adoption of BCT makes the retailer more inclined to target knowledgeable consumers, as indicated by $\bar{\phi}_0 < \phi_0$. This observation is

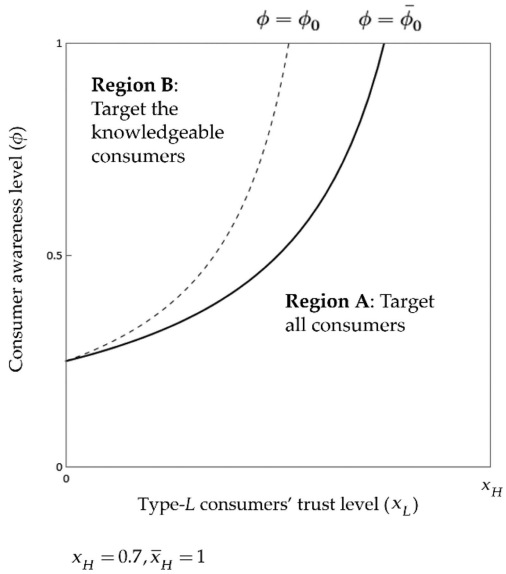
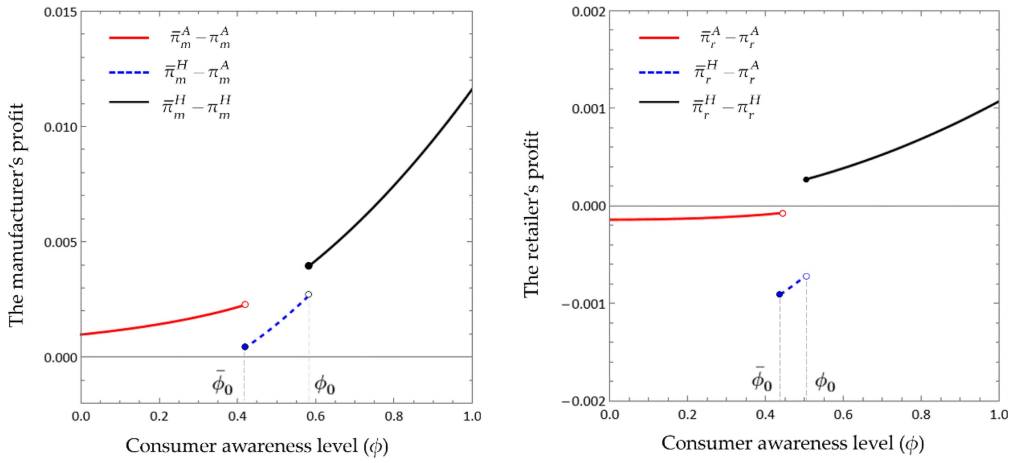


Fig. 4. Targeted consumer group in the two scenarios.

driven by the following factors. The adoption of BCT inherently benefits all consumers, including both type-H and type-L consumers, which helps attract a larger volume of demand. However, the adoption of BCT enhances the trust level of type-H consumers (i.e., $\bar{x}_H > x_H$), while the trust level of type-L consumers remains unchanged. Type-H consumers, who are knowledgeable about the benefits of BCT-enabled traceability, have a higher trust level in product quality. When the retailer's strategy targets type-H consumers, a high retail price is set due to the relatively high awareness level. This leads to a significant increase in the retailer's profit margin and effectively compensates for the BCT investment cost. Consequently, as shown in Fig. 4, region B becomes larger with the adoption of BCT (i.e., $\phi = \bar{\phi}_0$), indicating that the retailer is more inclined to target knowledgeable consumers compared with all consumers.

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(a): $\alpha = 0.2, k_q = 0.5, k_e = 1, \delta = 0.3, \gamma = 0.5, \bar{x}_H = 1, x_H = 0.7, x_L = 0.4$

(b): $\alpha = 0.2, k_q = 0.5, k_e = 0.5, \delta = 0.8, \gamma = 0.2, \bar{x}_H = 0.7, x_H = 0.6, x_L = 0.3$

Fig. 5. Impact of BCT adoption on supply chain members' performance.

C. Impact of BCT Adoption

BCT adoption may affect the profit of the manufacturer and retailer and then consequently the supply chain performance. We compare the supply chain members' profits between scenarios B and E to investigate the impact of BCT adoption. The key findings of this analysis are summarized in the following proposition.

Proposition 6: Impact of BCT adoption on supply chain members. (a) BCT adoption always improves the performance of the manufacturer, whereas it can lead to either an improvement or decline in the performance of the retailer, largely depending on the improvement in consumer trust level. (b) When $\gamma > \max\{\frac{1}{3} - \frac{k_e(7+16\alpha)}{36k_q(1+\alpha)\delta^2}, \frac{1}{3} - \frac{4k_e}{9k_q\delta^2}(-1 + \frac{256x_H^2(\bar{x}_H - x_L)^4 x_L^2}{\bar{x}_H^4(-4x_H x_L + \bar{x}_H(3x_H + x_L))^2(1+\alpha)})\}$, *BCT adoption always leads to a win–win situation for the supply chain members.*

Proposition 6(a) suggests that the manufacturer consistently benefits from the adoption of BCT, whereas the retailer may be worse off. As shown in Fig. 5(a), BCT adoption leads to a better performance of the manufacturer regardless of the targeted consumer group. Compared with the retailer, the manufacturer has the first-mover advantage, that is, it sets a wholesale price and product quality level first. This advantage allows the manufacturer to set a high wholesale price to offset the shared BCT investment cost. In an extreme situation where the manufacturer bears all the BCT investment cost, it can still set a high wholesale price to optimize its profit. In addition, the adoption of BCT inherently benefits all consumers, thereby helping to attract a larger volume of demand. Consequently, the manufacturer always benefits from BCT adoption regardless of the targeted group.

On the other hand, as shown in Fig. 5(b), BCT adoption may lead to a decline in the retailer's performance. As discussed in Proposition 3, the retailer's profit decreases with the sensitivity coefficient δ when the cost-sharing coefficient γ is small. In a situation with a high δ and a small γ, supply chain members make a substantial investment in BCT, with most of the investment cost borne by the retailer. However, the retailer's optimal retail price cannot cover its BCT investment cost because setting a substantially high retail price will offset the positive effect of BCT investment. Consequently, the retailer may be worse off with BCT adoption. In addition, we find that the adoption of BCT leads to higher overall supply chain profits. The benefits of BCT adoption include the improved trust level of knowledgeable consumers, increased consumers' willingness to pay, and elimination of retailer transaction costs. These advantages effectively offset the associated BCT investment cost, resulting in higher overall supply chain profits, regardless of the targeted consumer group.

BCT adoption leads to improved performance of the supply chain. Hence, the key lies in how to allocate profit between supply chain members to align their incentives for adopting BCT. Proposition 6(b) characterizes the condition under which BCT adoption leads to a win–win situation for both the manufacturer and the retailer. That is, both the manufacturer and retailer are incentivized to invest in BCT to improve supply chain traceability. In this situation, the BCT investment cost borne by the retailer is so small that the benefits from adopting BCT compensate for the shared BCT investment cost. Hence, a cost-sharing contract should be carefully designed to ensure that BCT can be adopted within the supply chain and that all supply chain members benefit. In practice, this finding helps the supply chain managers to design a feasible cost-sharing contract to encourage both the upstream and downstream enterprises to join the BCT adoption project.

Next, we examine the impact of BCT adoption on supply chain performance and have the following proposition.

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Proposition 7: BCT adoption makes the supply chain performance more sensitive to the consumer awareness level, namely, $\frac{\partial(\bar{\pi}_m+\bar{\pi}_r)}{\partial\phi} \geq \frac{\partial(\pi_m+\pi_r)}{\partial\phi}$.

Proposition 7 reveals an interesting impact of BCT adoption on supply chain performance. To be specific, adopting BCT makes the performance of the supply chain more sensitive to consumer awareness [33], [34]. In other words, we have $\frac{\partial(\bar{\pi}_m+\bar{\pi}_r)}{\partial\phi} \geq \frac{\partial(\pi_m+\pi_r)}{\partial\phi}$. According to the consumer's utility function $\theta(q+\delta e)\bar{x}_i - P$, adopting BCT improves the trust level of knowledgeable consumers, and the improved supply chain traceability inherently benefits consumers, thereby increasing consumers' willingness to pay. In addition, adopting BCT eliminates the retailer's transaction costs, which weakens the double marginalization effect in the supply chain. In this situation, an improvement in the level of consumer awareness can lead to a more substantial increase in the profit of the supply chain. Hence, from the perspective of the supply chain, it is important to improve the level of consumer traceability awareness after adopting BCT-enabled systems. In practice, it is more important for the supply chain members to promote the benefits of BCT after adopting BCT, which can raise the consumer traceability awareness level and more effectively improve the supply chain performance.

V. Discussion

A. Consumer Welfare and Social Welfare

In this section, we examine the impact of BCT adoption on consumer welfare and social welfare. In scenario B, we denote consumer welfare (social welfare) as CW^H (SW^H) when the retailer targets knowledgeable consumers and CW^A (SW^A) when the retailer targets all consumers. Consumer welfare can be measured by consumer surplus, which is the sum of net utilities for all consumers [39] and can be specified as follows:

$$\mathrm{CW}^H = \phi \int_{\frac{P}{qx_H}}^{1} (\theta q x_H - P)\, d\theta$$

$$\mathrm{CW}^A = \phi \int_{\frac{P}{qx_H}}^{1} (\theta q x_H - P)\, d\theta + (1-\phi) \int_{\frac{P}{qx_L}}^{1} (\theta q x_L - P)\, d\theta.$$

Hence, we have the expressions of social welfare [40] as follows:

$$\mathrm{SW}^H = \pi_m^H + \pi_r^H + \mathrm{CW}^H$$
$$\mathrm{SW}^A = \pi_m^A + \pi_r^A + \mathrm{CW}^A.$$

In scenario E with BCT adoption, we denote consumer welfare (social welfare) as $\overline{\mathrm{CW}}^H$ ($\overline{\mathrm{SW}}^H$) when the retailer targets the knowledgeable consumers and $\overline{\mathrm{CW}}^A$ ($\overline{\mathrm{SW}}^A$) when the retailer targets all consumers. Based on the economic definition of consumer welfare and social welfare, we have the expressions as follows:

$$\overline{\mathrm{CW}}^H = \phi \int_{\frac{P}{(q+\delta e)\bar{x}_H}}^{1} (\theta (q+\delta e)\, \bar{x}_H - P)\, d\theta$$

$$\overline{\mathrm{CW}}^A = \phi \int_{\frac{P}{(q+\delta e)\bar{x}_H}}^{1} (\theta (q+\delta e)\, \bar{x}_H - P)\, d\theta + (1-\phi) \int_{\frac{P}{(q+\delta e)x_L}}^{1} (\theta (q+\delta e)\, x_L - P)\, d\theta$$

$$\overline{\mathrm{SW}}^H = \bar{\pi}_m^H + \bar{\pi}_r^H + \overline{\mathrm{CW}}^H$$
$$\overline{\mathrm{SW}}^A = \bar{\pi}_m^A + \bar{\pi}_r^A + \overline{\mathrm{CW}}^A.$$

The equilibrium consumer welfare is solved and summarized in Table A-I, and the social welfare can be derived accordingly. An interesting question is whether adopting BCT will benefit consumers. We investigate the impact of BCT adoption on consumer welfare and the results are summarized in the following proposition.

Proposition 8: Adopting BCT does not always improve consumer welfare. However, BCT adoption can improve the consumer welfare and social welfare with an identical targeted consumer group, namely, $\overline{\mathrm{CW}}^A \geq \mathrm{CW}^A$, $\overline{\mathrm{CW}}^H \geq \mathrm{CW}^H$, $\overline{\mathrm{SW}}^A \geq \mathrm{SW}^A$, and $\overline{\mathrm{SW}}^H \geq \mathrm{SW}^H$.

As Proposition 8 states, adopting BCT can always improve consumer welfare and social welfare when the targeted consumer group is identical. As shown in Fig. 6, when $\phi \in [0, \bar{\phi}_0]$ or $\phi \in [\phi_0, 1]$, indicating that the targeted consumer group is identical in both scenarios, the consumer welfare and social welfare benefits from the adoption of BCT. This is because traceability enabled by BCT is inherently beneficial to consumers and directly increases consumer utility. Also, the improved trust level further increases the utility of knowledgeable consumers. In addition, the elimination of transaction costs can help reduce the retail price. These changes brought about by adopting BCT consequently offset the BCT investment cost and attract more demand [3], [7], [36]. Hence, consumer welfare and social welfare increased after adopting BCT. Interestingly, when $\phi \in (\bar{\phi}_0, \phi_0)$, indicating that the targeted consumer group differs between scenarios B and E, adopting BCT can lead to a decline in consumer welfare. In this situation, adopting BCT with all consumers as the targeted group results in lower consumer welfare and social welfare. This is because the benefits brought by BCT push up the retail price [we have $\bar{P}^A > P^H$ when $\overline{\mathrm{CW}}^A < \mathrm{CW}^H$ in Fig. 6(a)], which greatly decreases the demand volume, even though the retailer targets all consumers. However, the situation without BCT results in a lower retail price, even though the targeted group is only knowledgeable consumers, and the lower retail price attracts more demand. Therefore, in this situation, adopting BCT counterintuitively hurts consumer welfare and consequently social welfare.

B. Adopting BCT to Improve Consumers' Trust Level

One may assume that there is a direct impact of BCT adoption on improving consumers' trust level rather than consumer utility. In this section, we revise our model by capturing the direct and positive impact of BCT adoption on consumers' trust levels. Based on scenario E, a consumer's net utility can be rewritten as $\theta q(\bar{x}_i + \tau e) - P$, in which τ denotes the sensitivity degree of consumers' trust level to supply chain traceability. Intuitively, a large investment in BCT brings better supply chain traceability, which makes the consumers have a higher trust level in the

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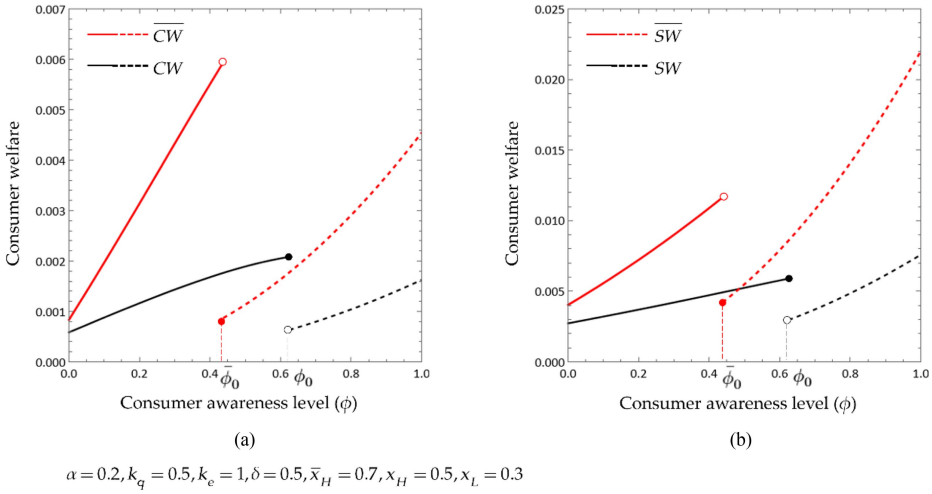


Fig. 6. Impact of awareness level on consumer welfare and social welfare. (a) Consumer welfare. (b) Social welfare.

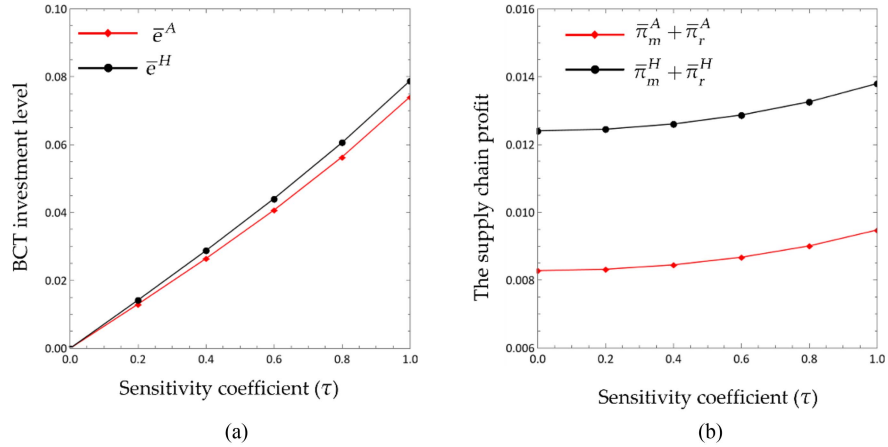


Fig. 7. Impact of sensitivity coefficient τ on the equilibrium.

products. Hence, considering the BCT adoption to improve consumers' trust level, the demand quantity is specified as follows:

$$Q\ (P,e) = \phi\left(1 - \frac{P}{q\,(\bar{x}_H + \tau e)}\right)^+ + (1-\phi)\left(1 - \frac{P}{q\,(x_L + \tau e)}\right)^+.$$

Similar to scenario E, we solve the equilibrium by backward induction. However, when the retailer targets all consumers, the equilibrium is too complicated to derive some analytical results. Hence, we conduct a numerical study to explore the impact of the sensitivity coefficient τ on the equilibrium outcome. The numerical results are graphically presented in Fig. 7, based on which we can derive several observations. First, a high sensitivity coefficient τ leads to a high BCT investment level, regardless of the targeted consumer group. Intuitively, a high sensitivity coefficient τ improves the consumers' trust level and amplifies the positive impact of BCT-enabled traceability, which drives the supply chain members to make a large BCT investment. Second, a high sensitivity coefficient τ leads to a better performance of the supply chain, regardless of the targeted consumer group. A high sensitivity coefficient τ results in a high trust level of the consumers, which helps improve the demand quantity. Although there is an increased BCT investment cost, a high consumer trust level consequently benefits the supply chain profits. However, both the manufacturer and retailer benefit depends on the design of the cost-sharing contract. To be sure, we can develop a feasible cost-sharing contract for the benefit of all the supply chain members.

VI. Conclusion

A. Concluding Remarks and Major Findings

BCT has been widely employed as a promising solution for ensuring supply chain traceability recently. In this article, we explored the impacts of consumer awareness on the decisions and performance of supply chain members under ERP-enabled and BCT-enabled traceability. In scenario B, we find that a high level of awareness serves to elevate the equilibrium wholesale

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price, retail price, and product quality, subsequently leading to a substantial increase in the profits of supply chain members. In addition, a low level of awareness prompts the retailer to target all consumers, whereas a high level of awareness shifts its focus to knowledgeable consumers. In scenario E, we show that the equilibrium outcomes, including the BCT investment level, always increase with the level of awareness and consumer trust. In addition, we demonstrate that adopting BCT always leads to improved performance for the manufacturer, whereas it can lead to either improved or diminished performance for the retailer.

Moreover, we examine the impact of adopting BCT and show that it makes the retailer more inclined to target knowledgeable consumers. Interestingly, we show that adopting BCT makes the performance of the supply chain more sensitive to the level of consumer awareness. This finding suggests that the supply chain members promote the benefits of BCT after adopting BCT because it can raise the consumer traceability awareness level and more effectively improve the supply chain performance. Finally, we examine the impact of adopting BCT on consumer welfare and social welfare. We find that adopting BCT can improve consumer welfare when the targeted consumer group is identical. However, adopting BCT is not always beneficial to consumer welfare and social welfare.

B. Managerial Implications and Future Study

The results have several potential managerial implications. First, in practice, consumer traceability awareness greatly affects the adoption of BCT [4]. Our study develops a stylize model to the supply chain managers to measure the impact of consumer traceability awareness on BCT adoption in supply chains. Second, we provide a feasible cost-sharing contract [see Proposition 6(b)] for the upstream and downstream supply chain members to consider BCT adoption. Such a cost-sharing contract can help ensure that all the supply chain members benefit from the BCT adoption, which is the participation constraint of supply chain members when considering BCT adoption. Third, the supply chain members are suggested to promote the benefits of BCT to raise the consumer traceability awareness level and trust level. We show that the increase of awareness level and consumer trust level always has positive impacts on the performance of the supply chain members. Finally, our results provide operational strategies for supply chain members to adopt BCT considering consumer traceability awareness. For example, we suggest that the retailer targets all consumers when the awareness level is low and shifts its focus solely to the knowledgeable consumers when the awareness level is high.

Our model has certain limitations, and we raise several issues and avenues for future research. First, in the blockchain platform, the number of linked nodes and the degree of disclosed information will affect the supply chain members' decisions. Future research can explore the impact of BCT adoption by capturing more specific characteristics of BCT, such as information disclosure in the network. Second, our findings are derived from a stylized model that focuses on key factors to maintain tractability. To validate the robustness of our analytical findings, future studies could employ simulation-based or empirical analyses.

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